<u>Hints and Formulas</u>

Percent Applications

$change = new - initial$	$\% \, of \; Change = \frac{Change}{Initial} * 100$
$Percent: Proportion$ $\frac{is}{of} = \frac{\%}{100}$	$Percent: Equation$ $is \rightarrow equal \; to$ $of \rightarrow \; multiply$ $what \rightarrow variable$
$Sale \; Price = Initial \; Price - Discount$	$\% \, Discount = \frac{Discount}{Initial \; Price} * 100$
$Profit = Selling \; Price - Cost \; Price$	$\% \, Profit = \frac{Profit}{Cost \; Price} * 100$
$Loss = Cost \; Price - Selling \; Price$	$\% \, Loss = \frac{Loss}{Cost \; Price} * 100$
If A is $p\%$ more than B, then $A = (100 + p)\%$ of B	
If A is $p\%$ less than B, then $A = (100 - p)\%$ of B	

Index Numbers

$$\frac{price \; in \; \$Y}{price \; in \; \$X} = \frac{CPI_Y}{CPI_X}$$

$$rate\ of\ inflation = \frac{CPI\ value\ of\ final - CPI\ value\ of\ initial}{CPI\ value\ of\ initial} * 100$$

$$\frac{price\ (Town\ 2)}{price\ (Town\ 1)} = \frac{index\ (Town\ 2)}{index\ (Town\ 1)}$$

$$index\ number = \frac{value}{reference\ value} * 100$$

Mathematics of Finance Literacy

$$Monthly\ interest\ payment = Monthly\ interest\ rate * Average\ balance$$

$$Net\ monthly\ cash\ flow = Monthly\ income - Monthly\ expenses$$

$$\begin{aligned} &Market\ capitalization \\ &\qquad = Total\ Number\ of\ Outstanding\ Shares \\ &\qquad * Current\ Share\ Price \end{aligned}$$

$$Total\ Return = \frac{A - P}{P} * 100\%$$

Total Return = ((A – P)/P) * 100%

$$Annual\ Return = \left[\left(\frac{A}{P}\right)^{\frac{1}{t}} - 1\right] * 100\%$$

Annual Return = ((A/P)^(1/t) – 1) * 100%

Simple Interest

$SI\ = \ Prt$ $\qquad SI = A - P$

$$SI = \frac{Art}{1 + rt}$$

SI = (A * r * t)/(1 + r * t)

$$P = \frac{SI}{rt}$$

P = SI / (r * t)

$$P = \frac{A}{1 + rt}$$

P = A / (1 + r * t)

$$t = \frac{SI}{Pr}$$

t = SI / (P * r)

$$t = \frac{A - P}{Pr}$$

t = (A – P) / (P * r)

$$r = \frac{SI}{Pt}$$

r = SI / (P * t)

$$r = \frac{A - P}{Pt}$$

r = (A – P) / (P * t)

$$A = P(1 + rt)$$

$$A = P + SI$$

Compound Interest

$$A = P\left(1+\frac{r}{m}\right)^{mt}$$

A = P * (1 + (r / m))^(m * t)

$$A = P + CI \qquad CI = A - P$$

$$P = \frac{A}{\left(1+\frac{r}{m}\right)^{mt}}$$

P = A / ((1 + (r / m))^(m * t))

$$r = m * \left[\left(\frac{A}{P}\right)^{\frac{1}{mt}} - 1\right]$$

r = m * ((A / P)^(1 / (m * t)) – 1)

$$r = m * \left(10^{\frac{\log\left(\frac{A}{P}\right)}{mt}} - 1\right)$$

r = m * (10^(log (A / P)) / (m * t)) – 1)

$$t = \frac{\log\left(\frac{A}{P}\right)}{m\log\left(1+\frac{r}{m}\right)}$$

t = (log (A / P)) / (m * log (1 + (r / m)))

If Compounded:	$m =$
Annually	1 (1 time per year: once every year)
Semiannually	2 (2 times per year: every six months)
Quarterly	4 (4 times per year: every three months)
Monthly	12 (12 times per year: every month)
Weekly	52 (52 times per year: every week)
Daily (Ordinary)	360 (360 times per year: banker's year)
Daily (Exact)	365 (365 times per year: every day)

Continuous Compound Interest

$$A = Pe^{rt}$$

$$P = \frac{A}{e^{rt}}$$

P = A / (e^(r * t))

$$t = \frac{\ln\left(\frac{A}{P}\right)}{r}$$

t = (ln (A / P)) / r

$$r = \frac{\ln\left(\frac{A}{P}\right)}{t}$$

r = (ln (A / P)) / t

$$CCI = A - P$$

Annual Percentage Yield (APY)

<u>APY for Compound Interest</u>

$$APY = \left(1 + \frac{r}{m}\right)^m - 1$$

APY = (1 + (r / m))^m – 1

$$r = m\left[(APY+1)^{\frac{1}{m}} - 1\right]$$

$$r = m\left(\sqrt[m]{APY+1} - 1\right)$$

r = m * ((APY + 1)^(1 / m) – 1)

APY for Continuous Compound Interest

$$APY = e^{r} - 1$$

$$r = \ln[APY + 1]$$

Future Value of Ordinary Annuity and Sinking Fund

$$FV = m * PMT * \left[\frac{\left(1+\frac{r}{m}\right)^{mt} - 1}{r}\right]$$

FV = m * PMT * (1 / r) * ((1 + (r / m))^(m * t) – 1)

$$Total\ PMTs = PMT * m * t$$

$$CI = FV - (PMT * m * t) \quad or \quad CI = FV - Total\ PMTs$$

$$t = \frac{\log\left[\frac{r * FV}{m * PMT} + 1\right]}{m * \log\left(1 + \frac{r}{m}\right)}$$

t = log(((r * FV) / (m * PMT)) + 1) * (1 / m) * (1 / (log(1 + (r / m))))

$$PMT = \frac{r * FV}{m * \left[\left(1 + \frac{r}{m}\right)^{mt} - 1\right]}$$

PMT = (r * FV) / (m * ((1 + (r / m))^(m * t) – 1))

Present Value of Ordinary Annuity and Amortization

$$PV = m * PMT * \left[\frac{1-\left(1+\frac{r}{m}\right)^{-mt}}{r}\right]$$

PV = m * PMT * (1 / r) * (1 – (1 + (r / m))^(-1 * m * t))

$$Total\ PMTs = PMT * m * t$$

$$CI = PMT * m * t - PV \quad or \quad CI = Total\ PMTs - PV$$

$$t = -\frac{\log\left[1 - \frac{r * PV}{m * PMT}\right]}{m * \log\left(1 + \frac{r}{m}\right)}$$

t = -1 * (1 / m) * (1 / (log (1 + (r / m)))) * log(1 – ((r * PV) / (m * PMT)))

$$PMT = \frac{PV}{m} * \left[\frac{r}{1 - \left(1 + \frac{r}{m}\right)^{-mt}}\right]$$

PMT = (PV / m) * (r / (1 – (1 + (r / m))^(-m * t)))